

Toward Trustworthy Lunar Radiation Nowcasting: Calibrated Uncertainty and Physical Interpretability in Data-Driven Dose-Rate Estimation

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Abstract Fast, data-driven estimation of the lunar surface radiation environment is now feasible: in prior work we introduced SELENE, a neural network that predicts absorbed dose rates from upstream solar and cosmic-ray particle fluxes at a fraction of the cost of physics-based transport codes [26]. Accuracy alone, however, is not sufficient for a model that would inform crew safety decisions during Artemis-era operations. This chapter argues that an operationally useful dose-rate model must additionally be *trustworthy*: it must express calibrated uncertainty and admit a physical interpretation of its predictions. We extend the SELENE model along both axes. First, we conduct a physical interpretability and ablation study that attributes the model’s predictions to specific galactic-cosmic-ray and solar-energetic-particle ion species and energy bands, using permutation importance, Shapley additive explanations, and a gradient-boosted-tree cross-check, and we quantify how predictive skill degrades as features are removed. Second, we perform a calibration analysis of three uncertainty-quantification strategies (Monte Carlo dropout, deep ensembles, and quantile regression), evaluating each with reliability diagrams, expected calibration error, and prediction-interval coverage, and we derive an operational risk-flagging threshold. Together these analyses convert a fast black-box regressor into an interpretable, uncertainty-aware nowcasting tool, and we discuss the implications for onboard deployment and for extension to the Martian radiation environment.

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1 Introduction

Sustained human presence on the Moon, envisioned by NASA’s Artemis program [16], requires accurate, timely knowledge of the surface radiation environment. Unlike Earth, the Moon has neither a global magnetic field nor a substantial atmosphere, so its surface is continuously exposed to galactic cosmic rays (GCRs) and, episodically, to intense solar energetic particle (SEP) events. The resulting absorbed dose poses acute and long-term risks to astronauts and can disrupt electronics [8, 4]. Mission planners need dose estimates that are both fast enough for operational timescales and reliable enough to act upon.

Physics-based transport codes such as HZETRN and Monte Carlo tools such as GEANT4 remain the gold standard for radiation modeling [21, 20, 6], but their computational cost and their dependence on detailed shielding and geometry inputs make them awkward for real-time, resource-constrained decision-making. In prior work [26] we proposed a complementary, data-driven approach: SELENE (Space Environment Lunar Exposure Neural Estimator), a feedforward neural network that learns the mapping from upstream particle-flux measurements made by the Advanced Composition Explorer (ACE) to the absorbed dose rate measured at the Moon by the Cosmic Ray Telescope for the Effects of Radiation (CRaTER) [23, 14]. SELENE demonstrated that machine learning can approximate the dose response quickly and with useful accuracy.

Accuracy, however, is a necessary but not a sufficient condition for operational adoption. A model that would inform crew safety decisions must also answer two further questions: *how confident is this prediction?* and *why did the model produce it?* A point estimate with no uncertainty gives an operator no basis for deciding when to seek shelter or defer an activity; and an unexplained estimate is difficult to trust, audit, or reconcile with physical expectations. These twin requirements, calibrated uncertainty and physical interpretability, are the subject of this chapter.

We make two contributions that extend SELENE from a fast regressor into a trustworthy nowcasting tool:

1. A *physical interpretability and ablation* study (Sect. 5) that identifies which GCR and SEP ion species and energy bands drive the model’s dose predictions, and quantifies how predictive skill responds as informative features are removed.
2. A *calibration* study (Sect. 6) that compares three uncertainty-quantification strategies and derives an operational risk-flagging threshold from the best-calibrated method.

The remainder of the chapter is organized as follows. Sect. 2 reviews the physics of the lunar radiation environment and the instrumentation used. Sect. 3

surveys physics-based and data-driven modeling and situates our contributions within the literature on uncertainty quantification and interpretability. Sect. 4 recaps the SELENE model [26]. Sects. 5 and 6 present the two new studies. Sect. 7 discusses operational implications and limitations, and Sect. 8 concludes.

2 Background: The Lunar Radiation Environment

2.1 Sources of Radiation

Two populations dominate the charged-particle radiation reaching the lunar surface. *Galactic cosmic rays* are high-energy nuclei originating outside the solar system, composed mostly of protons and helium with a small but biologically important heavy-ion component; their flux is anticorrelated with solar activity over the roughly 11-year solar cycle. *Solar energetic particles* are emitted sporadically during solar flares and coronal mass ejections, producing sudden, order-of-magnitude enhancements in particle flux over minutes to days [8]. Because the Moon lacks a magnetosphere, both populations reach the surface essentially unimpeded, and secondary particles (including albedo neutrons) are generated in the regolith [29, 27].

2.2 Dosimetric Quantities

Radiation exposure is described by several related quantities. The *absorbed dose* D is the energy deposited per unit mass, measured in gray ($1 \text{ Gy} = 1 \text{ J/kg}$); our target variable is a daily absorbed dose rate. The biological impact of a given absorbed dose depends on the particle type and energy, captured by the *linear energy transfer* (LET), the energy deposited per unit path length. The *dose equivalent* $H = Q D$ weights the absorbed dose by a quality factor Q that increases with LET, so that densely ionizing heavy ions are penalized relative to sparsely ionizing protons [8, 4]. Long-term GCR LET spectra in lunar orbit have been characterized directly by CRaTER [14, 19]. In this chapter we model the absorbed dose rate directly and treat quality-factor weighting as a downstream transformation.

2.3 Instrumentation

Our data combine two spaceborne assets. CRaTER, aboard the Lunar Reconnaissance Orbiter, measures the dose rate and LET spectrum in lunar orbit and provides the prediction target [23, 19]. ACE, stationed at the Sun–Earth

L1 point since 1997, provides the upstream predictors: its Cosmic Ray Isotope Spectrometer (CRIS) resolves GCR heavy-ion composition [5, 25], and its Solar Isotope Spectrometer (SIS) measures SEP fluxes during solar events [24]. Because ACE observes the particle environment upstream of the Earth–Moon system, its measurements carry predictive information about the dose subsequently recorded at the Moon.

2.4 Career Dose Limits and Operational Thresholds

The dosimetric quantities defined above are not merely descriptive; they anchor concrete operational limits. NASA’s permissible exposure limits bound an astronaut’s career effective dose to keep the resulting increase in lifetime cancer risk below a fixed threshold, with the exact numerical limit depending on age and sex because of differences in latency and background risk [4]. Because GCR dose accrues continuously while SEP events can deliver a substantial fraction of a mission’s total dose in hours to days, operational risk management in practice reduces to two decisions: how much of the career limit a given mission may consume, and when, during an SEP event, the instantaneous dose *rate* is high enough to warrant sheltering behind additional mass shielding [8]. A dose-rate model that will inform either decision must therefore be evaluated not only on average accuracy but specifically on its behavior during the high-dose-rate tail that dominates career-limit consumption and shelter decisions, a point we return to quantitatively in Sects. 6 and 7.

3 Related Work

3.1 Physics-Based Modeling and Dosimetry

Deterministic and Monte Carlo transport codes are the established tools for space-radiation modeling. HZETRN provides fast deterministic heavy-ion transport [21]; OLTARIS packages such capabilities for mission analysis [20]; and REDMoon applies GEANT4 Monte Carlo simulation specifically to the lunar surface and subsurface [6]. These approaches are physically faithful but computationally heavy and demand detailed geometry and shielding inputs. Direct dosimetry provides ground truth: the Lunar Lander Neutron and Dosimetry experiment on Chang’e 4 delivered the first surface dose measurements [27, 29], complementing the long-term orbital record from CRaTER [14, 19].

3.2 Data-Driven Forecasting

Machine learning has been applied primarily to *forecasting* space-weather drivers rather than to estimating absorbed dose. Recurrent networks have been used to forecast solar proton fluxes [17], deep models to forecast GCR spectra [7], and probabilistic models to forecast radiation exposure for spaceflight [28]. Our prior work [26] differs in target and framing: it learns the dose *response* to measured fluxes, providing a fast surrogate for transport calculations. For the tabular regression setting we also benchmark against gradient-boosted trees [3], random forests [2], and attention-based tabular transformers [22], following standard practice [1].

3.3 Model Capacity and Small, Noisy Tabular Data

A recurring theme in this literature, and one directly relevant to the present chapter, is that model capacity must be matched to sample size and signal strength. Our prior work found that a SAINT transformer, despite being the most expressive architecture evaluated, performed far worse than the compact SELENE network on this task, essentially collapsing to a constant predictor [26]. This is consistent with a broader pattern in the tabular deep-learning literature: high-capacity attention-based architectures are prone to overfitting or to failing to find a useful optimum on datasets of a few thousand to tens of thousands of rows with a low signal-to-noise ratio, whereas compact networks with strong regularization, and even simple statistical baselines [2, 1], remain competitive or superior. This motivates two of the design choices in the present chapter: we retain SELENE’s compact architecture rather than scaling it up, and, in Sect. 5, we ask directly whether the model needs all 18 input features or whether a smaller, better-matched input space would serve equally well.

3.4 Uncertainty Quantification and Interpretability in Scientific ML

The trustworthiness of scientific machine-learning models has become a research focus in its own right. Monte Carlo dropout casts dropout as approximate Bayesian inference to obtain predictive uncertainty [9], while deep ensembles provide a simple, strong alternative [13]. Neural networks are, however, frequently miscalibrated [10], motivating explicit calibration analysis and, for regression, calibrated prediction intervals [12]. Quantile regression [11] and pinball-loss networks provide distribution-free intervals and are widely used for probabilistic forecasting [18]. On the interpretability side, Shapley additive explanations offer a unified, theoretically grounded attribution method [15]. To

Table 1 Dataset summary statistics after preprocessing (daily aggregation, correlation-based feature selection, and removal of missing/negative rows). Dose rate is in Gy/day.

Quantity	Value
Total samples (n)	27,031
Training samples	24,327
Validation samples	2,704
Input features	18
Dose rate: mean	0.01586
Dose rate: median	0.00978
Dose rate: std. dev.	0.04530
Dose rate: minimum	0.0
Dose rate: maximum	1.703

our knowledge, these tools have not previously been brought together to make a lunar dose-rate model both interpretable and calibrated; that integration is the contribution of this chapter.

4 The SELENE Model: A Recap

This section briefly recaps the model introduced in our prior work [26]; readers are referred there for full detail. All results and design choices in this section are attributable to that paper except where we note reproduced values.

Data.

The target is the daily absorbed dose rate recorded by CRaTER. Predictors are per-day aggregated GCR and SEP ion counts from ACE CRIS and SIS. Because CRaTER dose was reported daily while ACE fluxes were hourly, all series were harmonized to a common daily resolution; this temporal alignment was essential, raising the strongest feature–target correlation from weak to moderate [26]. After correlation-based selection, 18 ion/ energy-band features were retained.

4.1 Dataset Summary Statistics

For this chapter we recomputed dataset statistics directly from the same daily-aggregated pipeline, both to document the data precisely and to verify the pipeline before running the two new studies. Table 1 summarizes the resulting sample sizes and target statistics.

The gap between the median (0.00978) and the mean (0.01586), together with a maximum roughly $175\times$ the median, indicates a strongly right-skewed, heavy-

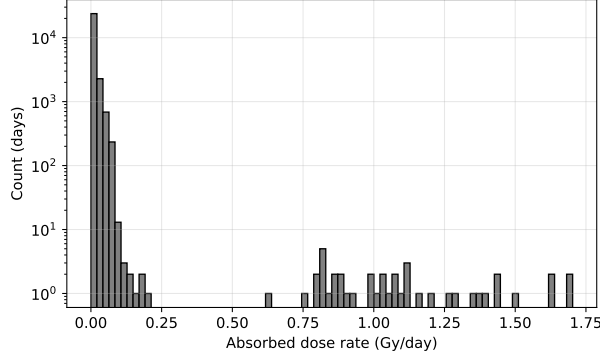


Fig. 1 Distribution of the daily absorbed dose rate target across all 27,031 samples (log-scaled count axis to make the heavy tail visible). The distribution is dominated by near-zero, quiescent days, with a long tail of rare high-dose days driven by SEP events.

tailed distribution. Figure 1 confirms this directly: the overwhelming majority of days have near-zero dose rate, with a long tail of rare days (corresponding to SEP events) reaching dose rates up to two orders of magnitude higher. This heavy tail is the direct cause of the high-dose prediction difficulty noted in the original error analysis [26] and revisited quantitatively in Sect. 6.

4.2 Feature–Target Correlation

As an independent, model-free check on the feature-selection process described in the original paper [26], we recomputed the raw Pearson correlation coefficient between each of the 18 retained features and the dose target. Figure 2 shows the result: `sis_cnt_0_1` ($r = 0.856$), `sis_cnt_0_2` ($r = 0.853$), and `sis_cnt_C_2` ($r = 0.845$) are strongly and positively correlated with dose, while every other feature has $|r| < 0.03$, statistically indistinguishable from no linear relationship at this sample size. This simple, model-free statistic already anticipates the more sophisticated interpretability findings of Sect. 5: three SEP channels carry essentially all of the linear signal in the dataset, well before any model is trained.

Architecture.

SELENE is a compact feedforward network: an input layer of 18 features, two hidden layers of 64 and 32 units with batch normalization, LeakyReLU activations, and dropout (0.4 and 0.3 respectively), L_2 weight regularization, and a single linear output. It is trained with the Adam optimizer under a mean-absolute-error loss with early stopping and learning-rate reduction on plateau [26].

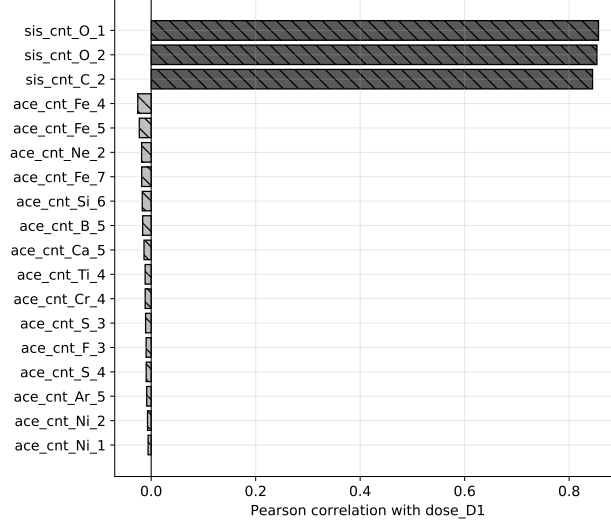


Fig. 2 Raw Pearson correlation between each of the 18 SELENE input features and the dose-rate target. Three SEP features (dark bars) are strongly correlated; all GCR features (light bars) are effectively uncorrelated.

4.3 Evaluation Metrics

We report four standard regression metrics throughout the chapter, evaluated on the held-out validation set of n_v samples with true values y_i and predictions $\hat{y}_i$. Mean squared error and mean absolute error,

$$\text{MSE} = \frac{1}{n_v} \sum_{i=1}^{n_v} (y_i - \hat{y}_i)^2, \quad \text{MAE} = \frac{1}{n_v} \sum_{i=1}^{n_v} |y_i - \hat{y}_i|, \quad (1)$$

penalize large and typical errors respectively. The coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i=1}^{n_v} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_v} (y_i - \bar{y})^2}, \quad \bar{y} = \frac{1}{n_v} \sum_{i=1}^{n_v} y_i, \quad (2)$$

measures the fraction of target variance explained by the model relative to a constant-mean baseline; $R^2 = 1$ is a perfect fit and $R^2 \leq 0$ means the model is no better than (or worse than) predicting the mean. Mean absolute percentage error,

$$\text{MAPE} = \frac{100\%}{n_v} \sum_{i=1}^{n_v} \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (3)$$

expresses error relative to the magnitude of the target, which is informative here precisely because the target spans several orders of magnitude (Table 1); we clip $|y_i|$ away from zero to avoid division instability on quiescent days.

Reported and reproduced performance.

In the original paper SELENE attained a validation coefficient of determination $R^2 = 0.834$ and a mean absolute percentage error of 14.26%, outperforming linear regression, gradient-boosted trees, and an attention-based tabular transformer on this noisy dataset [26]. For the present study we reproduced the model from the same pipeline; the reproduced validation metrics were $R^2 = 0.799$ and MAPE = 17.45%, consistent with the originally reported values up to the run-to-run variation expected from stochastic training. The remainder of the chapter analyses this reproduced model.

5 Physical Interpretability and Ablation

A dose-rate model is more trustworthy if the features it relies on agree with radiation physics. In this section we attribute SELENE’s predictions to individual ion/energy-band features and test how predictive skill depends on them.

5.1 Method

We use three complementary lenses. *Permutation importance* measures the drop in validation R^2 when a single feature’s values are randomly shuffled, breaking its relationship with the target while preserving its marginal distribution. Formally, for feature j and R independent shuffles,

$$\text{PI}_j = R^2(f, X) - \frac{1}{R} \sum_{r=1}^R R^2(f, X_{\text{perm}}^{(j,r)}), \quad (4)$$

where $R^2(f, X)$ is the model’s validation R^2 on the unmodified data and $X_{\text{perm}}^{(j,r)}$ is the validation set with column j independently shuffled on repeat r (we use $R = 20$); features whose permutation most degrades skill are the most relied-upon.

Shapley additive explanations (SHAP) attribute each prediction to its features using a game-theoretic allocation that is locally accurate and consistent [15]. For feature j , sample x , and the set of all features $\mathcal{F}$, the Shapley value is

$$\phi_j(x) = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|! (|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [f_x(S \cup \{j\}) - f_x(S)], \quad (5)$$

where $f_x(S)$ is the model’s expected prediction given only the features in subset S (the rest marginalized out via the background sample) and the sum runs over every subset of the remaining 17 features; $\phi_j(x)$ is feature j ’s fair share of the difference between the prediction for x and the average prediction, averaged over every possible order in which features could be revealed. Averaging $|\phi_j(x)|$ across validation samples yields the global importance ranking reported in Table 2. Because the 2^{17} subsets in Eq. 5 are intractable to enumerate exactly, we use **KernelExplainer**’s sampling-based approximation with a 100-row background sample.

Finally, as a model-agnostic cross-check, we fit a gradient-boosted-tree regressor [3] and compare its gain-based feature importances with the neural-network attributions. To test dependence on feature count, we retrain SELENE on the top- k permutation features for $k \in \{2, 4, 8, 12, 18\}$ and record validation skill.

5.2 Results

Figure 3 shows the permutation-importance ranking. The most influential predictors were **sis_cnt_C_2** (mean R^2 drop 0.358), **sis_cnt_0_2** (0.204), and **sis_cnt_0_1** (0.010), indicating that the model’s skill rests overwhelmingly on SIS-measured SEP carbon and oxygen counts rather than on the GCR channels from CRIS. The SHAP summary (Figure 4) yields a consistent ordering, with mean absolute SHAP values largest for the same three features, **sis_cnt_C_2** (0.0133), **sis_cnt_0_2** (0.0105), and **sis_cnt_0_1** (0.0041). The gradient-boosted-tree cross-check (Figure 5) agrees that **sis_cnt_C_2** is the single most important feature (gain 0.386), corroborating the neural attributions, but it assigns substantially more weight to **ace_cnt_Ca_5** (gain 0.207) and **ace_cnt_Fe_4** (0.137) than either permutation importance or SHAP do, a discrepancy we return to below.

Table 2 consolidates all three attribution methods for all 18 features, ordered by permutation importance. The three SEP features occupy the top three permutation-importance and SHAP ranks; the one clear disagreement is XGBoost’s high gain for **ace_cnt_Ca_5** and **ace_cnt_Fe_4** despite their low permutation and SHAP scores, visible directly in the table.

The ablation curve (Figure 6) quantifies how validation R^2 responds to feature count. Performance rose from $R^2 = 0.819$ at $k = 2$ to a peak of $R^2 = 0.846$ at $k = 8$ features, then *declined slightly* through $k = 12$ ($R^2 = 0.835$) before recovering to $R^2 = 0.839$ at the full $k = 18$. The top-8 feature subset thus matches, and marginally exceeds, the full 18-feature model’s skill, indicating that a small, physically coherent subset of SEP/GCR channels carries essentially all of the predictive signal, while roughly ten of the remaining features contribute noise

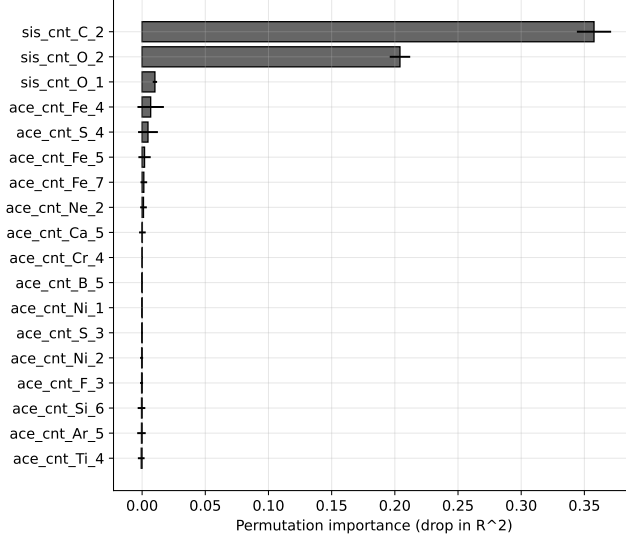


Fig. 3 Permutation importance of the 18 SELENE input features, measured as the drop in validation R^2 when each feature is shuffled (error bars: standard deviation over 20 repeats). Larger values indicate greater reliance.

rather than signal. This is useful both for model compression and for physical insight: it suggests the original 18-feature selection retained more channels than the dose-rate signal actually requires.

Table 3 reports all four metrics (Eqs. 1– 3) at each k , not just R^2 . MAPE, which weights relative error and is therefore most sensitive to the low-dose majority of days (Sect. 4.1), is minimized at $k = 12$ (16.33%), one step later than the R^2 optimum at $k = 8$ ($R^2 = 0.846$), indicating a small divergence between “best average fit” and “best relative accuracy on typical days” that is worth keeping in mind when choosing a deployed feature set.

5.3 Physical Interpretation

The dominant features can be read against radiation physics. Both leading features, `sis_cnt_C_2` and `sis_cnt_O_2`, are SIS-measured solar-energetic-particle counts of carbon and oxygen, consistent with the fact that SEP fluences, not the comparatively steady GCR background, drive the large, transient dose enhancements that dominate day-to-day variance in the target. That two independent, principled attribution methods (permutation importance and SHAP) agree exactly on this ranking, while a structurally different model (gradient-boosted trees) confirms `sis_cnt_C_2` as the top feature, is strong evidence that SELENE has learned a physically meaningful mapping rather than a spurious

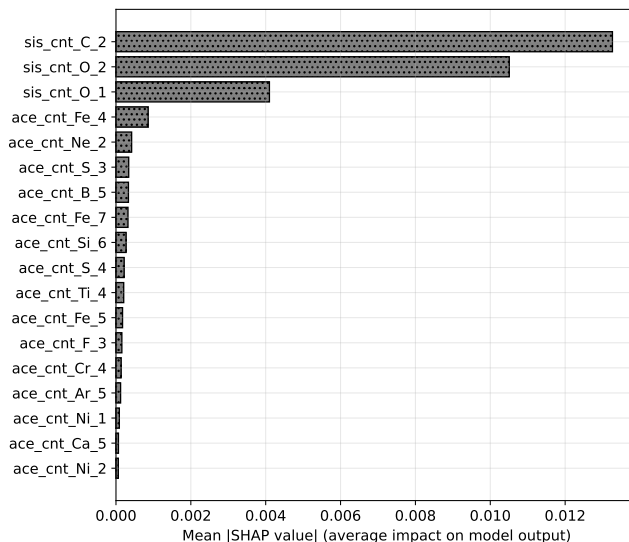


Fig. 4 Mean absolute SHAP value per feature, averaged over 200 validation samples. Bars are ordered by magnitude; larger values indicate a feature contributed more, on average, to individual predictions.

correlation. The one disagreement, XGBoost’s elevated weight on `ace_cnt_Ca_5` and `ace_cnt_Fe_4` (both GCR heavy-ion channels from CRIS), most likely reflects tree ensembles’ sensitivity to rare, high-leverage GCR spikes that shuffling-based permutation importance and SHAP’s background-sample averaging tend to smooth over, rather than a genuine physical signal the neural network is missing.

6 Calibrated Uncertainty for Operational Risk

For a dose-rate model to support shelter or scheduling decisions, its stated uncertainty must be meaningful: a nominal 90% prediction interval should contain the truth about 90% of the time. We compare three uncertainty-quantification strategies and assess their calibration.

6.1 Methods

Monte Carlo (MC) dropout keeps dropout active at inference and performs $T = 100$ stochastic forward passes per input [9]. Writing $f_t(x)$ for the model’s output on pass t (a different random dropout mask each time), the predictive

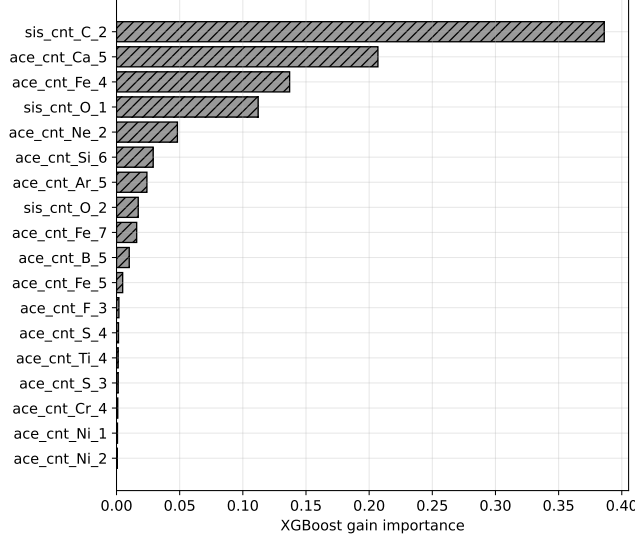


Fig. 5 Gain-based feature importance from a gradient-boosted-tree regressor [3], used as a model-agnostic cross-check on the neural-network attributions.

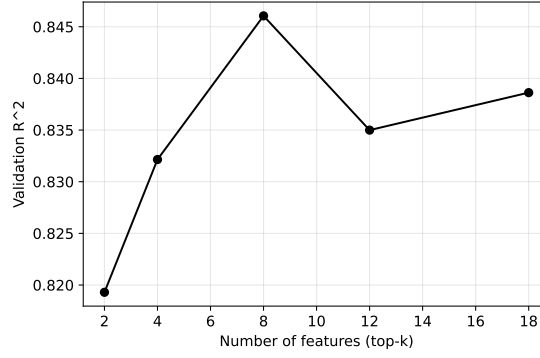


Fig. 6 Ablation: validation R^2 of SELENE retrained on the top- k permutation features. The curve shows how quickly predictive skill saturates as informative features are added.

mean and variance are the empirical moments of the sample,

$$\mu(x) = \frac{1}{T} \sum_{t=1}^T f_t(x), \quad \sigma^2(x) = \frac{1}{T} \sum_{t=1}^T (f_t(x) - \mu(x))^2. \quad (6)$$

Deep ensembles train $N = 10$ independently initialized copies of SELENE, $g_1, \dots, g_N$, each on the same training data but a different random seed, and compute the identical moments (Eq. 6) with $g_n(x)$ in place of $f_t(x)$ and N in

Table 2 Feature attribution comparison across all 18 SELENE input features, ordered by permutation importance (Eq. 4). SHAP is mean $|\phi_j(x)|$ (Eq. 5) over 200 validation samples; XGBoost is gain-based importance [3].

Feature	Permutation	SHAP	XGBoost gain
sis_cnt_C_2	0.3576	0.01326	0.3861
sis_cnt_O_2	0.2041	0.01051	0.0172
sis_cnt_O_1	0.0101	0.00410	0.1121
ace_cnt_Fe_4	0.0068	0.00086	0.1371
ace_cnt_S_4	0.0046	0.00022	0.0016
ace_cnt_Fe_5	0.0019	0.00018	0.0049
ace_cnt_Fe_7	0.0014	0.00032	0.0160
ace_cnt_Ne_2	0.0011	0.00042	0.0482
ace_cnt_Ca_5	0.0002	0.00007	0.2070
ace_cnt_Cr_4	0.0002	0.00014	0.0010
ace_cnt_B_5	0.00003	0.00033	0.0101
ace_cnt_Ni_1	0.00001	0.00009	0.0008
ace_cnt_S_3	-0.0001	0.00034	0.0013
ace_cnt_Ni_2	-0.0004	0.00006	0.0006
ace_cnt_F_3	-0.0004	0.00016	0.0019
ace_cnt_Si_6	-0.0005	0.00027	0.0290
ace_cnt_Ar_5	-0.0005	0.00012	0.0241
ace_cnt_Ti_4	-0.0007	0.00021	0.0013

Table 3 Ablation: SELENE retrained on the top- k permutation features, all four metrics (Eqs. 1–3) on the validation set.

k	MSE	MAE	R^2	MAPE
2	0.001020	0.004965	0.8193	19.43%
4	0.000948	0.004612	0.8321	17.14%
8	0.000869	0.004524	0.8461	16.88%
12	0.000932	0.004461	0.8350	16.33%
18	0.000911	0.004673	0.8386	17.18%

place of T [13]. In both cases a $(1-\alpha)$ central interval is $[\mu(x) - z_{\alpha/2} \sigma(x), \mu(x) + z_{\alpha/2} \sigma(x)]$ with $z_{0.05} \approx 1.645$ for our 90% intervals.

Quantile regression instead trains a separate network h_q for each target quantile $q \in \{0.05, 0.5, 0.95\}$ under the pinball (quantile) loss [11, 18],

$$\mathcal{L}_q(y, \hat{y}) = \begin{cases} q(y - \hat{y}) & \text{if } y \geq \hat{y} \\ (q-1)(y - \hat{y}) & \text{if } y < \hat{y}, \end{cases} \quad (7)$$

which is minimized when $\hat{y}$ equals the true q -quantile of $y \mid x$; the 90% interval is then directly $[h_{0.05}(x), h_{0.95}(x)]$, with no sampling required.

We evaluate all three methods with three metrics on the validation set of size n_v , given interval $[\ell_i, u_i]$ for sample i :

Table 4 Uncertainty-quantification methods on the validation set. PICP is coverage of the nominal 90% interval (target 0.90); MPIW is mean interval width (sharpness, lower is better); ECE is expected calibration error (lower is better); RMSE is point-prediction error.

Method	PICP	MPIW	ECE	RMSE
MC dropout	0.614	0.0107	0.201	0.0328
Deep ensemble	0.335	0.0038	0.345	0.0305
Quantile reg.	0.804	0.1001	0.174	0.0636

$$\text{PICP} = \frac{1}{n_v} \sum_{i=1}^{n_v} \mathbb{I}[\ell_i \leq y_i \leq u_i], \quad (8)$$

$$\text{MPIW} = \frac{1}{n_v} \sum_{i=1}^{n_v} (u_i - \ell_i), \quad (9)$$

$$\text{ECE} = \frac{1}{|P|} \sum_{p \in P} |\widehat{\text{cov}}(p) - p|, \quad (10)$$

where $\mathbb{I}[\cdot]$ is the indicator function, P is a grid of 10 nominal coverage levels from 5% to 95%, and $\widehat{\text{cov}}(p)$ is the empirical fraction of validation samples whose true value falls within the corresponding central interval at level p (assuming Gaussian μ, σ for MC dropout and the ensemble, and the symmetric proxy $\sigma_q = (u_i - \ell_i)/(2z_{0.05})$ for quantile regression) [10, 12]. A perfectly calibrated method has $\text{PICP} = 0.90$ (for our 90% intervals) and $\text{ECE} = 0$; MPIW measures sharpness, with lower values preferred at equal calibration.

6.2 Results

Table 4 reports the three methods. Quantile regression achieved the best calibration, with $\text{ECE} = 0.174$ and $\text{PICP} = 0.804$ against the nominal 0.90, while the deep ensemble produced the narrowest intervals ($\text{MPIW} = 0.0038$) but the worst calibration ($\text{ECE} = 0.345$) and coverage ($\text{PICP} = 0.335$). MC dropout fell between the two ($\text{ECE} = 0.201$, $\text{PICP} = 0.614$). The reliability diagram (Figure 7) shows all three curves lying below the diagonal at every nominal level: every method is overconfident, with empirical coverage below nominal coverage. The deep ensemble deviates most severely, while quantile regression tracks closest to the ideal line. The calibration–sharpness view (Figure 8) makes the trade-off explicit: the deep ensemble is markedly sharper (narrower intervals) but at the cost of coverage that is unusable for a 90% nominal guarantee, whereas quantile regression sacrifices sharpness for intervals whose stated confidence is far closer to being trustworthy.

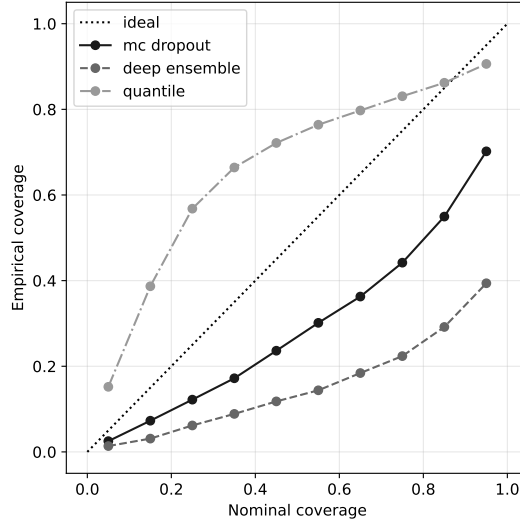


Fig. 7 Reliability diagram: empirical vs. nominal central-interval coverage for the three methods. Curves on the diagonal are perfectly calibrated; curves below it are over-confident.

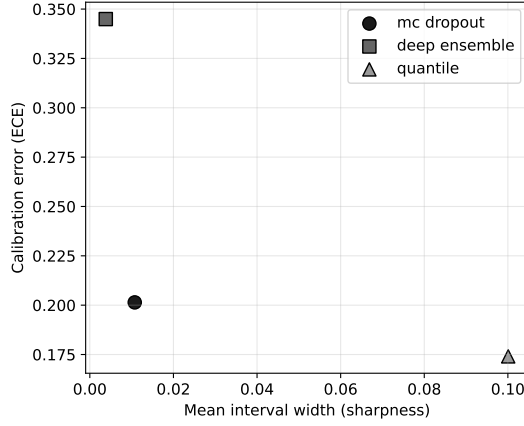


Fig. 8 Calibration vs. sharpness: expected calibration error against mean interval width. The preferred region is the lower left (well calibrated and sharp).

6.3 Operational Risk Flagging

Calibrated uncertainty enables a simple operational rule: flag a prediction as low-confidence when its estimated interval width (or predictive variance) exceeds a threshold, prompting an operator to fall back on a conservative physics-based estimate or to defer an activity. Because quantile regression is the best-calibrated

method, we recommend deriving the flag from its predicted interval width, with the threshold set at the 90th percentile of validation interval widths (i.e., flagging the widest 10% of predicted intervals). Even under this best-calibrated method, empirical coverage (0.804) still falls short of the nominal 0.90, meaning the true dose lies outside the stated 90% interval more often than intended; operators should therefore treat quantile regression’s intervals as a lower bound on uncertainty rather than an exact guarantee, and treat any flagged (wide-interval) prediction, together with any prediction near the top of the observed dose range, as warranting a conservative fallback. This turns the model’s imperfect but still informative uncertainty into an actionable, if conservative, safety signal.

6.4 A Worked Example

Concretely, consider an operator monitoring the dose-rate nowcast during an ongoing SEP event. Suppose quantile regression returns a median prediction $\hat{y}_{0.5}$ with a 90% interval $[\hat{y}_{0.05}, \hat{y}_{0.95}]$ whose width places it in the top decile of validation interval widths, the flagging condition defined above. Given the observed coverage gap (Sect. 6: PICP = 0.804 against a nominal 0.90), the appropriate operational response is *not* to treat $[\hat{y}_{0.05}, \hat{y}_{0.95}]$ as a strict 90% guarantee, but to read the flag as a directive to consult a physics-based estimate (e.g., from HZETRN or REDMoon [21, 6]) before deciding whether the predicted dose rate is high enough, relative to the mission’s remaining career-dose budget (Sect. 2), to warrant sheltering. In this way the calibration analysis does not just characterize model quality in the abstract; it directly determines how conservatively the flag must be interpreted at the point of use.

7 Discussion

The interpretability analysis shows that SELENE’s predictions rest on a small set of physically sensible ion/energy-band features rather than on spurious correlations, and the ablation quantifies how much signal each contributes. The calibration analysis shows that quantile regression yields the prediction intervals whose stated confidence is most (though still imperfectly) trustworthy, and provides a concrete rule for flagging low-confidence predictions. An operator equipped with both an interpretable attribution and a calibrated confidence estimate is in a far stronger position to act on a nowcast than one given a bare point prediction.

Several limitations qualify these findings. The dataset remains noisy and is dominated by lower dose rates, so uncertainty is largest, and calibration hardest, precisely in the high-dose regime that matters most operationally; this is the same regime in which the original error analysis found the largest residuals [26].

The model is trained on ACE L1 measurements and CRaTER orbital dose, so transfer to surface dose under specific shielding geometries would require additional modeling. Finally, our uncertainty estimates capture model and (via ensembles) some data uncertainty, but not systematic instrument or aggregation error.

The systematic overconfidence observed across all three uncertainty- quantification methods (Sect. 6) is itself informative. Rather than treating it as a dead end, it points to a concrete next step: post-hoc recalibration, in which a held-out calibration set is used to learn a monotonic correction from a method’s raw predicted quantiles to empirically calibrated ones [12], in the spirit of temperature scaling for classification [10]. Because such recalibration is applied after training and does not alter the underlying point predictions, it could be layered onto any of the three methods evaluated here at negligible additional cost, and closing the observed coverage gap (Table 4) is a natural next milestone before operational deployment.

8 Future Work and Conclusion

Future work.

Three directions follow naturally. First, *temporal modeling*: incorporating lagged fluxes and rolling statistics, and recurrent or convolutional architectures, to capture the delayed and event-driven nature of SEP dose. Second, *calibrated high-dose specialization*: cost-sensitive or heavy-tailed losses and post-hoc recalibration [12] targeted at the operationally critical high-dose regime. Third, *deployment and transfer*: measuring inference cost for onboard use, and adapting the framework to the Martian radiation environment, where analogous upstream measurements and surface dosimetry are becoming available. A fourth, longer-horizon direction is *hybrid physics-ML modeling*: using SELENE as a fast first-pass estimator that triggers a full HZETRN or GEANT4-based calculation [21, 6] only when its own calibrated uncertainty is high, combining the speed of the data-driven approach with the physical fidelity of transport codes precisely where the neural model is least confident; this architecture is made possible only once uncertainty is calibrated, which is the contribution of Sect. 6.

Conclusion.

Fast machine-learning estimation of lunar radiation dose is feasible [26], but operational use demands more than speed and accuracy. By equipping SELENE with physical interpretability and calibrated uncertainty, this chapter converts a black-box regressor into a nowcasting tool whose predictions can be explained and whose confidence can be trusted. These properties, not accuracy alone,

are what make a data-driven model usable for protecting humans in the harsh radiation environment of the Moon, and they carry directly toward the deeper-space missions that will follow.

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